Vraj Patel

ML Engineering | Applied NLP, Evaluation and ML Systems

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EXPERIENCE

Software Engineer, Internal Tools May 2025 - Present
CU Boulder | Institute of Behavioral Science

- Trained an unreleased BERT routing candidate on 6,137 tickets: 82.63% sealed-test accuracy and 0.7563 macro-F1, above the prior XGBoost baseline by 3.11 and 5.36 points.
- Keep model predictions advisory, with provenance, human review, separate training/serving paths, and runtime controls around external updates.
- Separate source intake from reportable workload and training eligibility; the dated 10,803-ticket snapshot yielded 6,487 eligible records across seven routing classes.

Lead Technical Research Assistant Mar 2023 - May 2025
CU Boulder | Institute of Behavioral Science

- Adapted Amazon Transcribe with a custom language model trained on about 110K Hindi words; WER fell from 40.1% to 30.1% on one held-out speech, not a corpus-wide benchmark.
- Built PyTorch/ResNet50 embedding and k-means analysis workflows with researchers studying visual bias in COVID-19 news imagery.

Undergraduate Research Assistant, Satellite Telemetry Aug 2024 - May 2025
The Data Mine of the Rockies | Purdue and L3Harris

- Trained Isolation Forest and LSTM models on synthetic NOS3 satellite telemetry to detect and classify simulated cyberattack anomalies.

Research Assistant, Data Engineering and ML Jan - Apr 2024
CU Boulder | Studio Lab

- Compared six Hindi speech-to-text engines with word-error-rate evaluation and assessed translation services; recommended Amazon Transcribe for the later IBS pipeline.

SELECTED PROJECTS

HateBlocker | Classical NLP Models
scikit-learn, XGBoost, TF-IDF

- Owned classical-ML exploration, 5,000-term TF-IDF features, 17 stylometric features, and grid-searched baselines in a five-person project; XGBoost reached 90.72% held-out accuracy.
- Evaluated three classes on 2,479 held-out examples; hateful-class recall remained 0.24. Teammates owned BERT, the FastAPI service, and browser extension.

Automated Interview Scoring
SentenceTransformers, scikit-learn, TensorFlow

- Team project using MiniLM alignment, lexical and 35 prosodic features, mutual-information selection, k-fold regressors, and SHAP analysis for 138 mock interviews.

EDUCATION

M.S. Computer Science + M.E. Engineering Management Expected May 2028
University of Colorado Boulder | Concurrent dual degree

B.S. Computer Science, Business Minor May 2025
University of Colorado Boulder | GPA: 3.84 technical, 3.65 overall

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, SQL, C++, Bash, PowerShell
ML: PyTorch, scikit-learn, Transformers, TensorFlow, NLP, model evaluation, SHAP
Data: PostgreSQL, SQLite, Redis, pgvector, Pandas, source reconciliation, data validation
Infrastructure: Docker, Kubernetes/OpenShift, AWS, GCP, Terraform, Linux, GitHub Actions